NagarSetu: Bilingual AI System for Civic Complaint and Information Assistance

Submitted in partial fulfilment of the requirements of the degree of

BACHELOR OF ENGINEERING

by

Prachiti Parab: 22106030

Aafreen Khan: 22106048

Sneha Gupta: 22106046

Tulsi Dubey: 22106055

Guide:

Prof. Rahul Ambekar

Department of Computer Science & Engineering

(Artificial Intelligence & Machine Learning)

A. P. SHAH INSTITUTE OF TECHNOLOGY, THANE

(2025-2026)

A. P. SHAH INSTITUTE OF TECHNOLOGY, THANE

CERTIFICATE

This is to certify that the project entitled **"NagarSetu: Bilingual AI System for Civic Complaint and Information Assistance"** is a bonafide work of **Prachiti Parab (**22106030**), Aafreen Khan (**22106048**), Sneha Gupta(**22106046**), Tulsi Dubey(**22106055**)** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering** in **Computer Science & Engineering (Artificial Intelligence & Machine Learning).**

Prof. Rahul Ambekar Guide	Prof. Sayali P. Badhan Project Coordinator
Prof. Dr. Jaya Gupta Head of Department	Dr. Uttam D. Kolekar Principal

A.P. SHAH INSTITUTE OF TECHNOLOGY, THANE

Project Report Approval for B.E.

This project report entitled **"NagarSetu: Bilingual AI System for Civic Complaint and Information Assistance"** by **"Prachiti Parab, Aafreen Khan, Sneha Gupta, Tulsi Dubey"** is approved for the degree of ***Bachelor of Engineering*** in ***Computer Science & Engineering (Artificial Intelligence & Machine Learning), 2025-26***.

Examiner Name	Signature
1._____	_____
2._____	_____

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Place:

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Prachiti Parab: 22106030

Aafreen Khan: 22106048

Sneha Gupta: 22106046

Tulsi Dubey: 22106055

Date

Abstract

The effective functioning of urban governance depends heavily on efficient grievance redressal systems and access to government-held information. Despite advancements in smart city technologies, civic complaints such as potholes, garbage accumulation, broken streetlights, or water leakage are often left unresolved due to inefficient reporting mechanisms. Similarly, the Right to Information (RTI) Act, a powerful tool for transparency, remains underutilized because citizens lack the awareness and ability to draft RTI requests in the correct format and language.

To address these challenges, this project introduces NagarSetu, a bilingual AI-powered system that bridges the gap between citizens and municipal authorities. NagarSetu enables citizens to file civic complaints through text, images, or voice inputs and provides AI-generated RTI drafts in Hindi and English. The system incorporates advanced technologies such as image classification using CNNs (MobileNet), NLP with BERT for complaint text categorization, Whisper for speech-to-text conversion, and RAG (Retrieval-Augmented Generation) for RTI drafting.

NagarSetu not only accelerates issue resolution but also strengthens democratic participation by empowering citizens to hold governance accountable through simplified RTI drafting. It aligns with multiple **Sustainable Development Goals (SDGs)** by enhancing inclusivity, resilience, transparency, and sustainability in urban management.

Keywords: CivicTech, Complaint Redressal, RTI, AI, NLP, Image Classification, Smart City, Governance, Bilingual Systems.

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Chapter 1
Introduction

Urban governance today faces numerous challenges in ensuring that civic issues reported by citizens are addressed efficiently and transparently. Cities are growing rapidly, and with this growth comes increasing pressure on municipal authorities to maintain infrastructure, sanitation, public safety, and essential services. Citizens frequently encounter problems such as **potholes, broken streetlights, uncollected garbage, water leakage, or other.** However, the process of reporting these issues and ensuring timely resolution is often slow and cumbersome. Traditional complaint management systems largely rely on manual submission methods, paperwork, and physical inspections. These methods are not only time-consuming but also prone to human error, miscommunication, and delays, which reduces citizen satisfaction and trust in local authorities.

The need for a more efficient system has become even more pressing with the concept of smart cities gaining traction. Smart cities aim to leverage technology to improve urban living, including better public services, transparent governance, and citizen participation. In this context, an intelligent platform capable of automating the registration, classification, routing, and monitoring of civic complaints can significantly enhance the responsiveness of municipal bodies. Such a platform can help bridge the gap between citizens and authorities, ensuring that grievances are addressed promptly and effectively.

The **NagarSetu system** has been developed to tackle these challenges. It is a multimodal, AI-driven platform that allows citizens to submit complaints through text, images, or voice recordings. By using advanced machine learning and natural language processing techniques, NagarSetu automatically classifies the complaints and routes them to the appropriate municipal department. This not only reduces manual effort but also ensures that complaints are handled by the right personnel, improving resolution speed and accuracy. Furthermore, the system includes features for generating **RTI (Right to Information) drafts,** enabling citizens to request information from government authorities efficiently, promoting transparency and accountability. Real-time notifications keep citizens updated about the status of their complaints, while geo-tagging ensures that authorities know the exact location of the reported issue. Additionally, administrators can utilize analytics dashboards and heatmaps to monitor trends, prioritize interventions, and allocate resources effectively.

1.1 Motivation

The primary motivation behind NagarSetu is the need to modernize and streamline civic complaint systems. Citizens frequently experience frustration when reporting common urban problems. For instance, a resident may notice a large pothole on a busy street that poses a safety hazard or a broken streetlight in a residential area that compromises safety at night. Currently, reporting such issues often involves multiple steps, such as filling out forms online, waiting in long queues at municipal offices, or calling different helplines. Even after submission, complaints may get misrouted or delayed due to manual classification or lack of coordination among departments.

By incorporating artificial intelligence, NagarSetu reduces human dependency in classifying complaints and ensures faster and more accurate routing to relevant authorities. This enhances citizen satisfaction and encourages more active participation in governance. Furthermore, the inclusion of RTI draft generation empowers citizens to demand accountability from government agencies.

1.2 Classification of Complaints

A key feature of NagarSetu is the image-based complaint classification module. Citizens can upload photos of civic issues, and the system automatically processes these images using Convolutional Neural Networks (CNNs), particularly MobileNet, a lightweight and efficient deep learning model. This module classifies complaints into distinct categories to ensure that they are routed to the correct municipal department for prompt action. The current categories include:

- **Potholes:** Damaged or broken roads that hinder traffic flow and pose safety risks to pedestrians and vehicles. For example, a deep pothole on a major street may lead to accidents if not addressed quickly.
- **Broken Streetlights:** Malfunctioning or non-functional streetlights that compromise safety at night, particularly in residential neighborhoods or busy intersections.
- **Garbage Dumping:** Illegal dumping or uncollected waste in public areas, creating unhygienic conditions, attracting pests, and affecting the environment.
- **Water Leakage:** Burst or leaking water pipelines that result in wastage of water, inconvenience to citizens, and potential damage to surrounding infrastructure.

Chapter 2

Literature Survey

Year	Title	Domain	Algorithm	Dataset	Gap
2024	Automated Complaint Classification& Routing Using NLP	NLP, ML	TF-IDF, Logistic Regression, BERT	Complaint datasets	No image classification
2025	AI-Powered Petition Analysis & Grievance Management System	AI/NLP, Civic governance	FastAPI backend, dashboards, ML for petitions	Citizen petition corpus	Limited RTI automation
2025	Digital Grievance Redressal for the Cleaner andSmarter India	Smart City, CivicTech	Location tagging + AI classification	Case study dataset	No bilingual/ voice support
2020	Complaint Management System using Mobile Apps for Smart Cities	Smart City, Mobile Systems	Android-based manual reporting	Public complaint forms	Manual entry, no AI
2019	Grievance Redressal through E-Governance – Government Pilot Study adverse drug reactions	E-Gov Platforms	Simple portals, no AI	Municipal datasets	Noautomation, poor scalability

2024	CitySolution: Mobile DL Complaint System	Deep Learning, Smart Cities	CNN-based image classification + monitoring	Android app complaint images	No RTI or voice input
2021	Computer Vision Assisted Approaches to Detect Street Garbage from Citizen Generated Imagery drug event detection	Computer Vision, Object Detection (CNN / Transfer Learning)	Traditional machine learning models, deep learning models	~2,500 images (with/without garbage), citizen-generated images from SF 311 service	Need for improving accuracy
2024	Pothole Detection Model using CNN and Metaheuristic Group Learning Algorithm	Computer Vision, Optimization	CNN + Metaheuristic Optimization	Public + collected pothole datasets	Limited to potholes; no complaint filing system
2023	Deep Learning-based Pothole Detection and Severity Classification with Location Mapping	Smart Cities, Computer Vision	CNN (Deep Learning) + Severity Classification	Road images (pothole vs. non-pothole)	No voice/RTI integration; only potholes

Chapter 3

Limitations of Existing Systems

Although several digital initiatives have been introduced for grievance redressal and e-governance, most remain inefficient and inaccessible. Existing systems are often unable to scale with the needs of rapidly growing urban populations, and they fail to provide a seamless citizen experience.

3.1 Language Barriers

Most civic complaint portals and RTI assistance systems are designed only in English. This excludes a large portion of the population, especially those in semi-urban and rural areas who are more comfortable with Hindi or regional languages. Even when bilingual support exists, it is often poorly implemented with limited translations, making the system confusing for first-time users.

Example: A pothole complaint portal in an Indian metro city was only available in English, which discouraged non-English-speaking citizens from reporting issues.

3.2 Manual and Tedious Processes

In many cities, citizens still need to:

- Fill lengthy online forms.
- Manually describe problems.
- Manually upload complaint documents.
- Wait for the municipal staff to sort, classify, and route complaints.

This not only delays resolution but also increases the chances of errors and lost records.

Case Study: A 2022 review of municipal complaint systems across Tier-1 Indian cities showed that average resolution times exceeded 60 days due to manual routing and classification of issues.

3.3 Lack of Multimedia Input

Modern smartphones allow citizens to easily capture images or record audio, but most complaint systems accept only text inputs. This limits citizens' ability to provide clear evidence of issues.

- A citizen reporting a garbage dump may be unable to show the actual condition through photos.
- Elderly or illiterate citizens may find voice input far more natural than typing, but this option is missing.

3.4 Weak Complaint Prioritization

Existing systems treat all complaints equally. A pothole on a small lane and an exposed live electric wire may both be given the same priority, leading to serious safety risks. Without AI-based prioritization, urgent issues are often buried under routine complaints.

3.5 Delayed RTI Assistance

Although the RTI Act empowers citizens, drafting a request requires:

- Knowledge of the correct format.
- Use of formal language.
- Awareness of the correct department.

Current online RTI portals merely act as submission gateways, with no guidance on drafting queries. This discourages ordinary citizens from exercising their rights to transparency.

3.6 Lack of Real-Time Tracking and Transparency

Once a complaint is filed, many platforms fail to provide live updates to citizens.

- Citizens are left wondering: "Has my complaint been assigned?" or "Is someone working on this?"
- Without tracking, trust in governance is reduced.

Some systems provide ticket numbers, but manual follow-ups with municipal offices are often required.

3.7 Absence of Community Verification

A common challenge is duplicate complaints (e.g., ten different people reporting the same pothole). Most platforms don't have a mechanism to:

- Merge similar complaints.
- Validate the authenticity of issues.

3.8 Weak Feedback Loops

After a complaint is marked as "resolved," citizens have no effective mechanism to verify the resolution or provide feedback.

- Issues may be closed on paper without real action.
- Citizens lose faith in complaint portals.

Chapter 4

Problem Statement, Objectives and Scope

4.1 Problem Statement

Urban citizens across India face recurring civic challenges such as potholes, broken streetlights, garbage accumulation, sewage leakage, and water supply disruptions. Despite the rise of digital governance, existing complaint platforms are plagued by:

- **Complex and unfriendly user interfaces**, which discourage citizens from reporting grievances.
- **Language barriers,** as most systems operate only in English, excluding large sections of society.
- **Manual complaint classification and routing**, leading to delayed resolutions.
- **Absence of multimedia inputs,** restricting users from submitting photos or voice complaints.
- **Limited RTI (Right to Information) assistance**, preventing effective citizen participation in governance transparency.

The consequence of these gaps is that complaints remain unresolved, RTI requests are underutilized, and trust in governance systems diminishes. therefore, there is a need for a bilingual, AI-powered, user-friendly, and scalable platform that empowers citizens to actively engage in governance through complaint reporting and RTI drafting.

4.2 Objectives

The project aims to design and develop NagarSetu, a bilingual AI system for civic complaint and information assistance. The specific objectives are:

1. **AI-enabled complaint reporting:** Enable complaint submission through text, images, and voice inputs and Use image classification (MobileNet) and NLP (BERT) for automatic complaint categorization.
2. **Geolocation integration:** Automatically detect complaint location using Google Maps API and Allow manual pin-drop for accuracy.
3. **Automated classification and routing:** Route complaints to the appropriate municipal department based on type and location and Introduce priority levels for high-risk issues such as live wires, water leakages, or garbage fires.
4. **RTI drafting support:** Simplify the Right to Information process by generating draft RTIs in English and Hindi. It Use a Retrieval-Augmented Generation (RAG) agent to provide contextually accurate drafts.
5. **Multi-role dashboard system:**

Citizens: File and track complaints, submit RTI requests.

Municipal staff: Receive, verify, and resolve issues.

Administrators: Oversee overall system performance and analytics.

Real-time tracking and notifications: Keep citizens informed about the status of their complaints through push notifications (Firebase, Twilio).

Community verification and feedback: Allow nearby residents to confirm or validate complaints and Collect feedback after resolution to ensure accountability.

4.3 Scope of the Project

The scope of NagarSetu includes:

Complaint Reporting

Citizens can submit complaints with images, text, and audio.

System integrates geotagging and assigns priority levels.

Automated Complaint Handling

AI modules classify and route complaints automatically.

Municipal staff receive structured issue reports for action.

RTI Assistance

Citizens can ask queries in Hindi/English, and the system generates RTI drafts in official format.

Drafts are downloadable as PDFs for submission.

Dashboards & Notifications

Real-time tracking for citizens.

Analytics dashboards for municipal staff and administrators and Push notifications for updates.

Community Features

Verification system to reduce duplicate or false complaints.

Feedback collection after issue resolution.

Sustainability Alignment

SDG 11.3: Inclusive, participatory, sustainable urban governance.

SDG 11.6: Sustainable waste management.

SDG 11.7: Prioritization of high-risk complaints for safety.

SDG 11.b: Strengthening community resilience via transparent processes.

Chapter 5

Proposed System

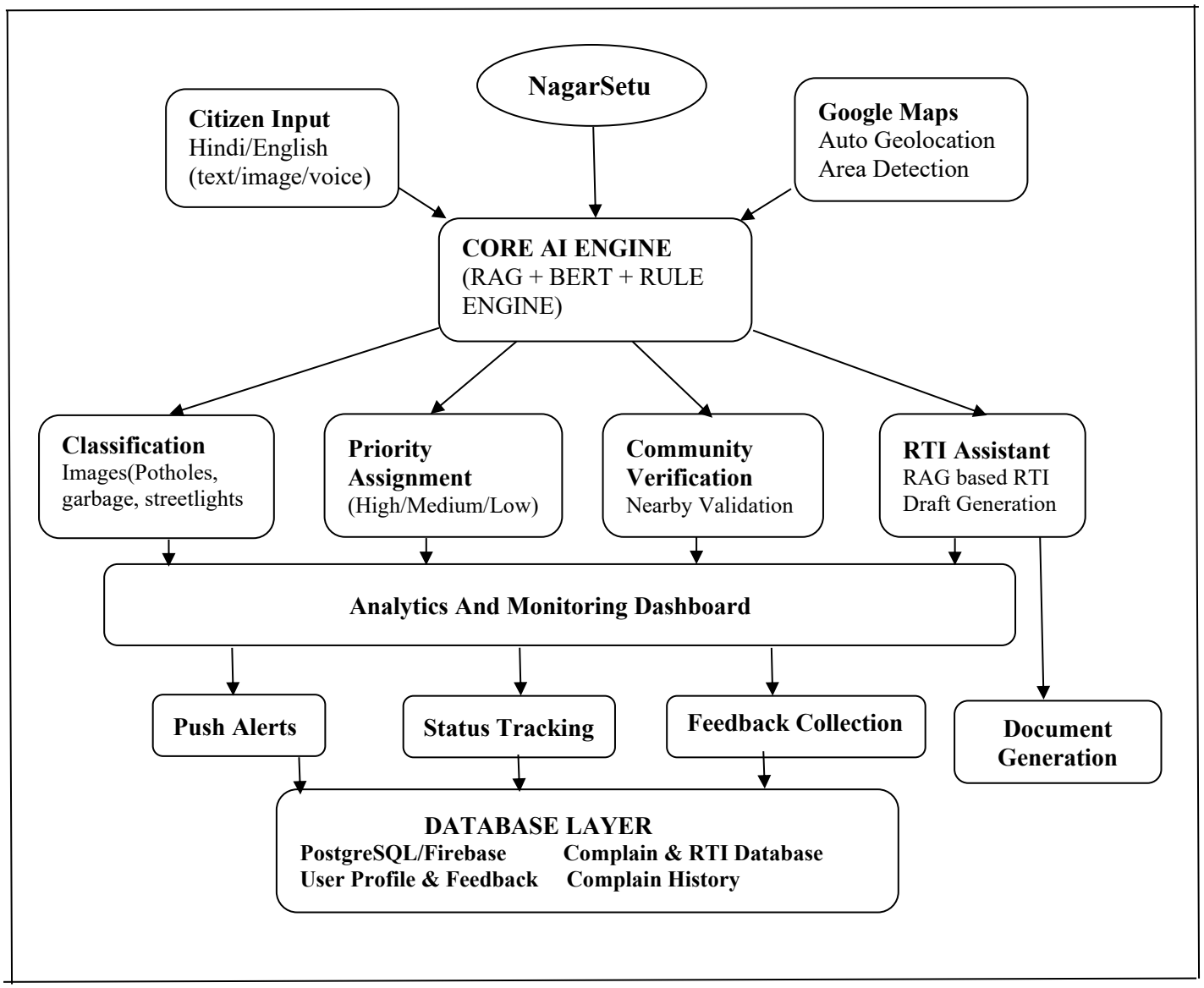


Figure 5.1 Architecture Diagram for NagarSetu

This diagram illustrates the workflow of an AI-powered civic management platform called NagarSetu. It visualizes how citizen inputs are processed through various intelligent components to streamline complaint management and RTI handling.

Key features include:

• **Citizen Input:** Users can submit complaints in Hindi or English via text, image, or voice.

• **Core AI Engine:** Combines RAG, BERT, and rule-based logic for multilingual understanding, continuous learning, priority assessment, and intelligent routing.

• **Google Maps Integration:** Enables automatic geolocation and area detection for precise complaint mapping.

• **Classification:** AI model categorizes issues like potholes, garbage, or water leakage using image recognition.

• **Priority Assignment:** Complaints are automatically ranked as High, Medium, or Low based on severity and context.

• **RTI Assistant:** RAG-based module generates RTI drafts automatically for transparency and accountability.

• **Community Verification:** Allows local users to validate complaints through nearby verification.

• **Analytics & Monitoring Dashboard:** Provides heatmaps, pattern recognition, complaint trends, and performance metrics for better decision-making.

• **Document Generation:** Produces RTI reports in PDF or Word format.

• **Push Alerts:** Sends emergency notifications to relevant authorities or citizens.

• **Feedback Collection:** Gathers citizen satisfaction data for service improvement.

• **Database Layer:** Utilizes PostgreSQL and Firebase for managing complaints, RTI data, notifications, and user profiles.

Data Flow Diagrams:

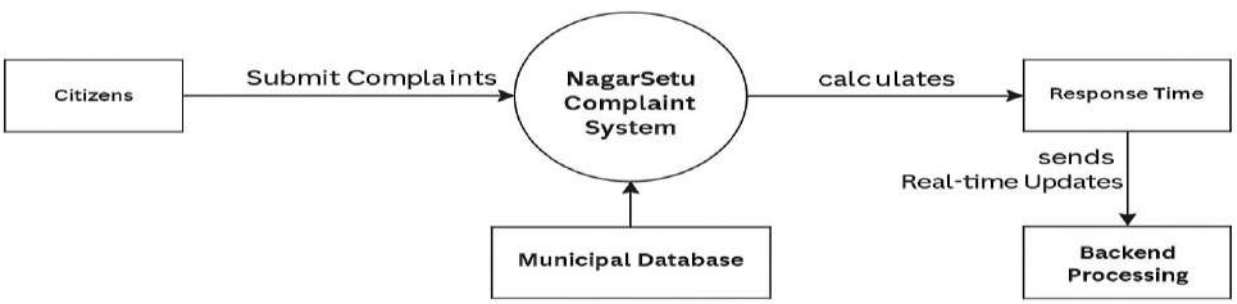


Figure 5.2 Level 0 diagram for NagarSetu

The NagarSetu Level 0 Data Flow Diagram provides a high-level view of how citizen inputs and system components interact. Citizens submit complaints or RTI requests via text, image, or voice. The CORE AI Engine (RAG + BERT + Rule Engine) processes these inputs, classifies issues, assigns priorities, and integrates with Google Maps for geolocation. Validations come from the community, while the Analytics Dashboard. Outputs such as alerts, documents, and feedback are shared with users and authorities, creating an efficient, transparent, and AI-driven civic management loop.

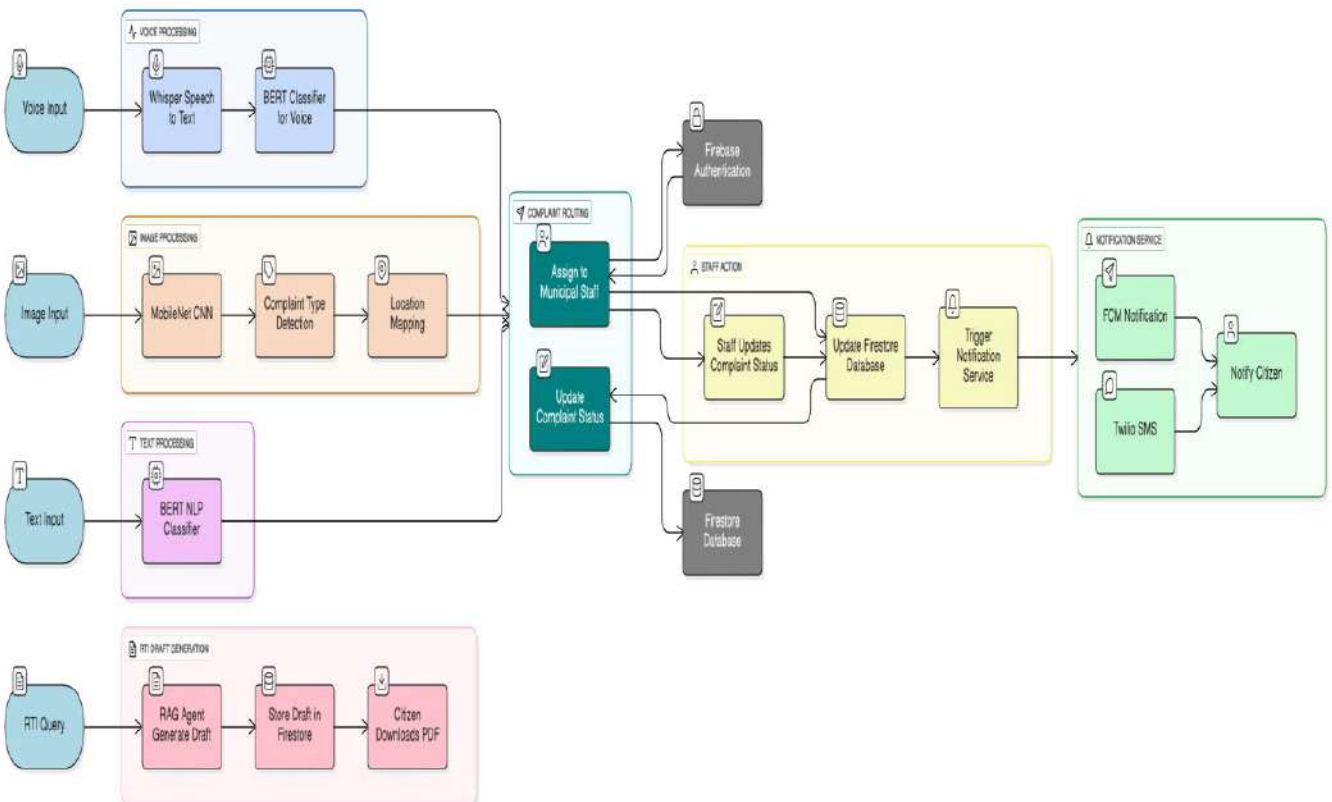


Figure 5.3 Level 1 diagram for Nagarsetu

The NagarSetu Level 1 Data Flow Diagram provides a detailed view of the system's internal processes and component interactions. It defines user roles such as citizens, administrators, and municipal officers, each with specific access and responsibilities. The diagram illustrates how citizen inputs (text, image, or voice) flow through the CORE AI Engine, which performs language detection, classification, and priority assignment using RAG, BERT, and rule-based logic.

It highlights key modules like Complaint Classification, Priority Assignment, RTI Draft Generation, Community Verification, and the Analytics Dashboard. Data from Google Maps and the database supports geolocation tagging, trend analysis, and performance tracking.

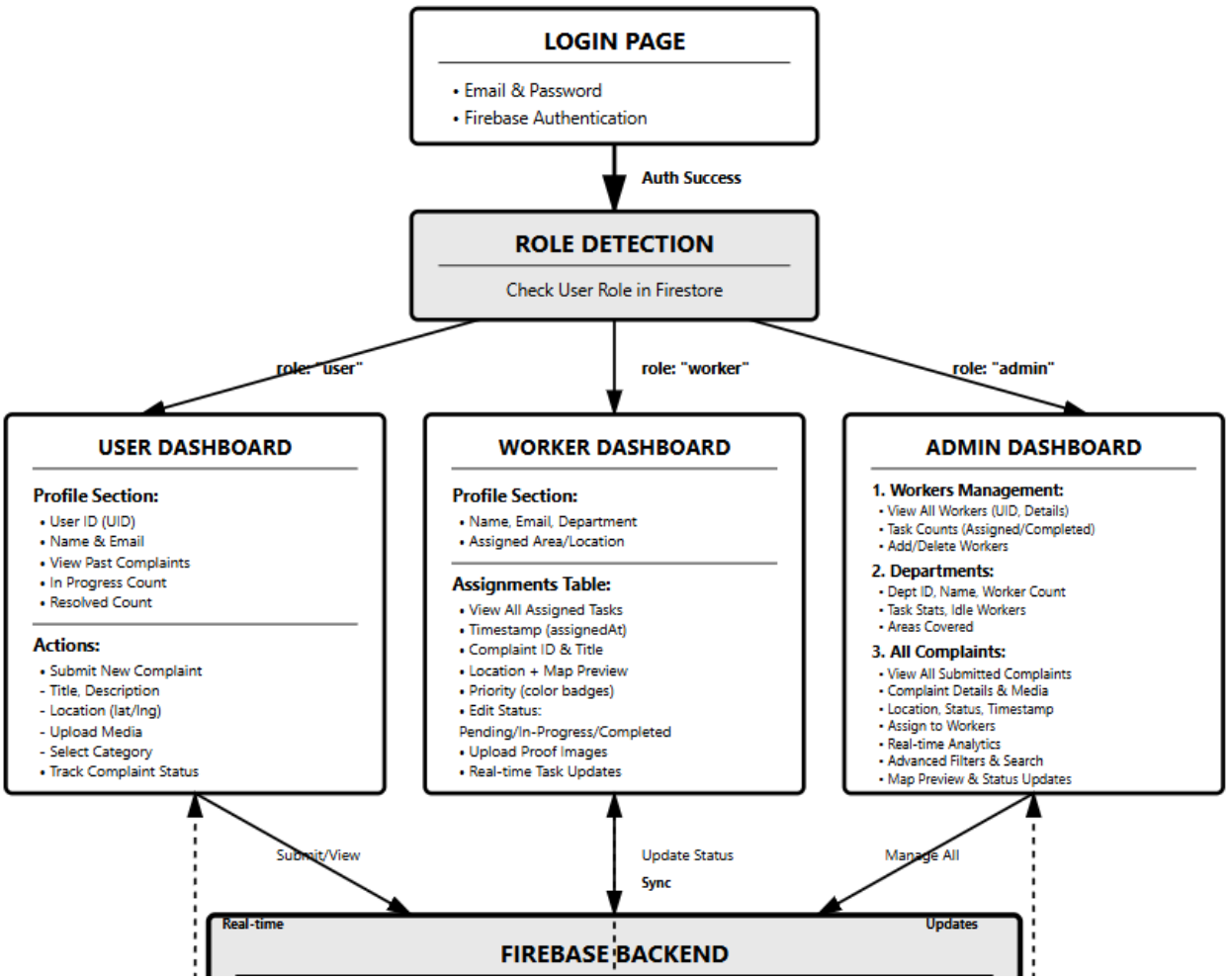


Figure 5.4 Level 2 diagram for NagarSetu

1**. Worker Dashboard:**

- Displays assigned complaints with ID, title, location (lat/lng), and priority.
- Allows status updates: Pending, In-Progress, Completed.
- Supports proof image uploads for task verification.

- Provides map preview of complaint location using SimpleMap.
- Shows worker profile with name, email, department, and operating area.

2. User (Citizen) Profile:

- Displays user details: UID, name, email.
- Shows complete complaint history with current status.
- Provides statistics and filters for In Progress and Resolved complaints.
- Enhances transparency and engagement by allowing users to track complaints in real time.

3. Admin Features:

- Manages workers, departments, and complaints efficiently.
- Tracks real-time task statistics: assigned, in-progress, completed.
- Supports adding/removing workers and toggling status.
- Enables department management: worker allocation, idle workers, total complaints, and completed tasks.
- Allows automatic complaint assignment to workers based on category and department.

4. Real-Time & System Features:

- Integrates Firebase Firestore for live updates with onSnapshot.
- Implements advanced filtering and search across all tables.
- Supports role-based authentication ensuring proper access control.
- Features interactive maps for location visualization.
- Provides file upload functionality and dynamic status management with notifications.
- Maintains a professional UI with badges, color coding, and clean design for better usability.

Use Case Diagram:

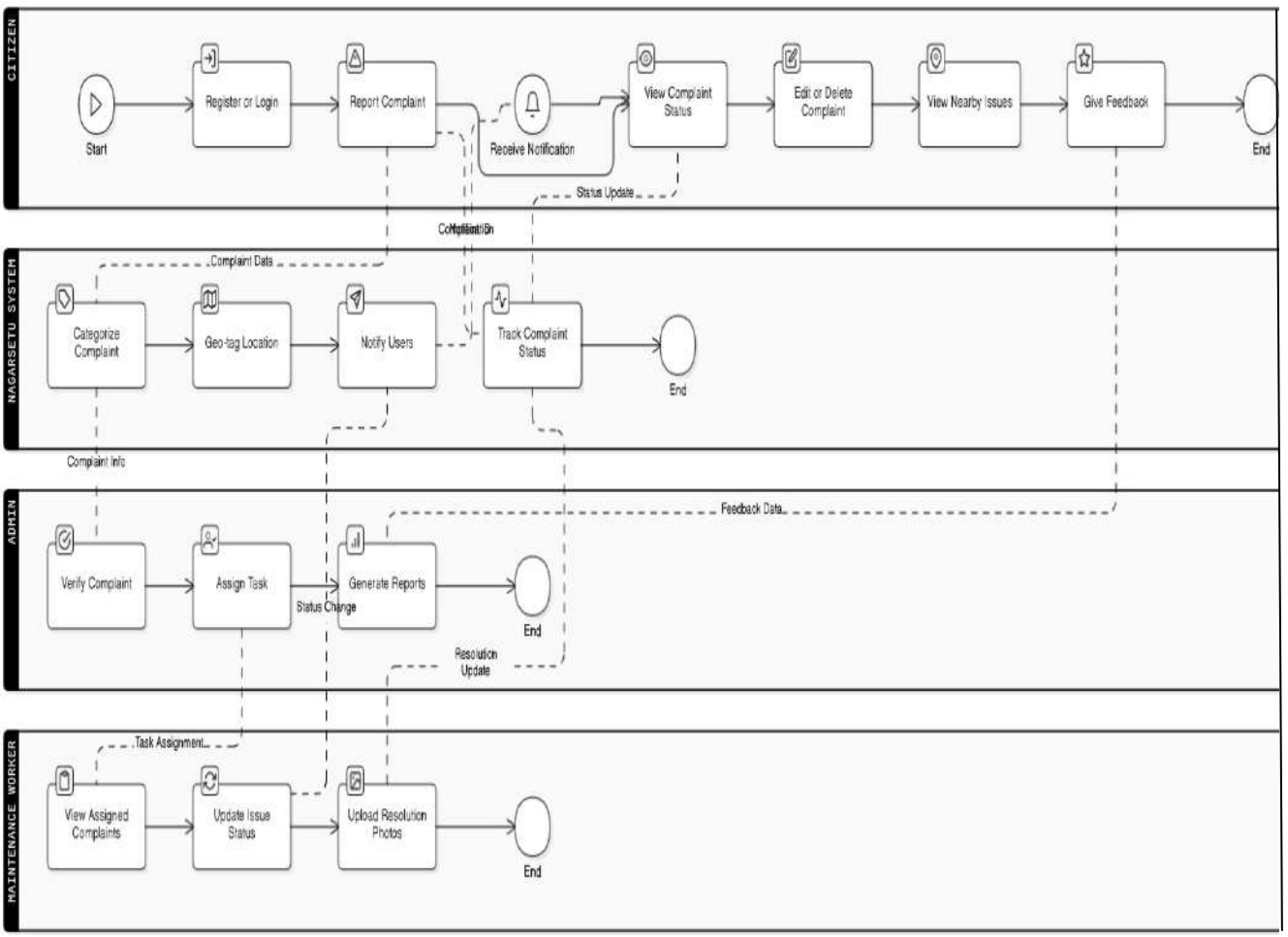


Figure 5.5 Use Case Diagram for NagarSetu

A Use Case Diagram is a UML diagram that shows the functional requirements of a system and how actors (users or other systems) interact with it. It includes actors, use cases, associations, and system boundaries to represent the system's scope. Use case diagrams help understand system behavior, plan development, and communicate with stakeholders. For example, in NagarSetu, actors like Citizen, Admin, and Maintenance Worker interact with use cases such as Register Complaint, Assign Complaint, and Resolve Issue, showing the flow from complaint submission to resolution.

Sequence Diagram:

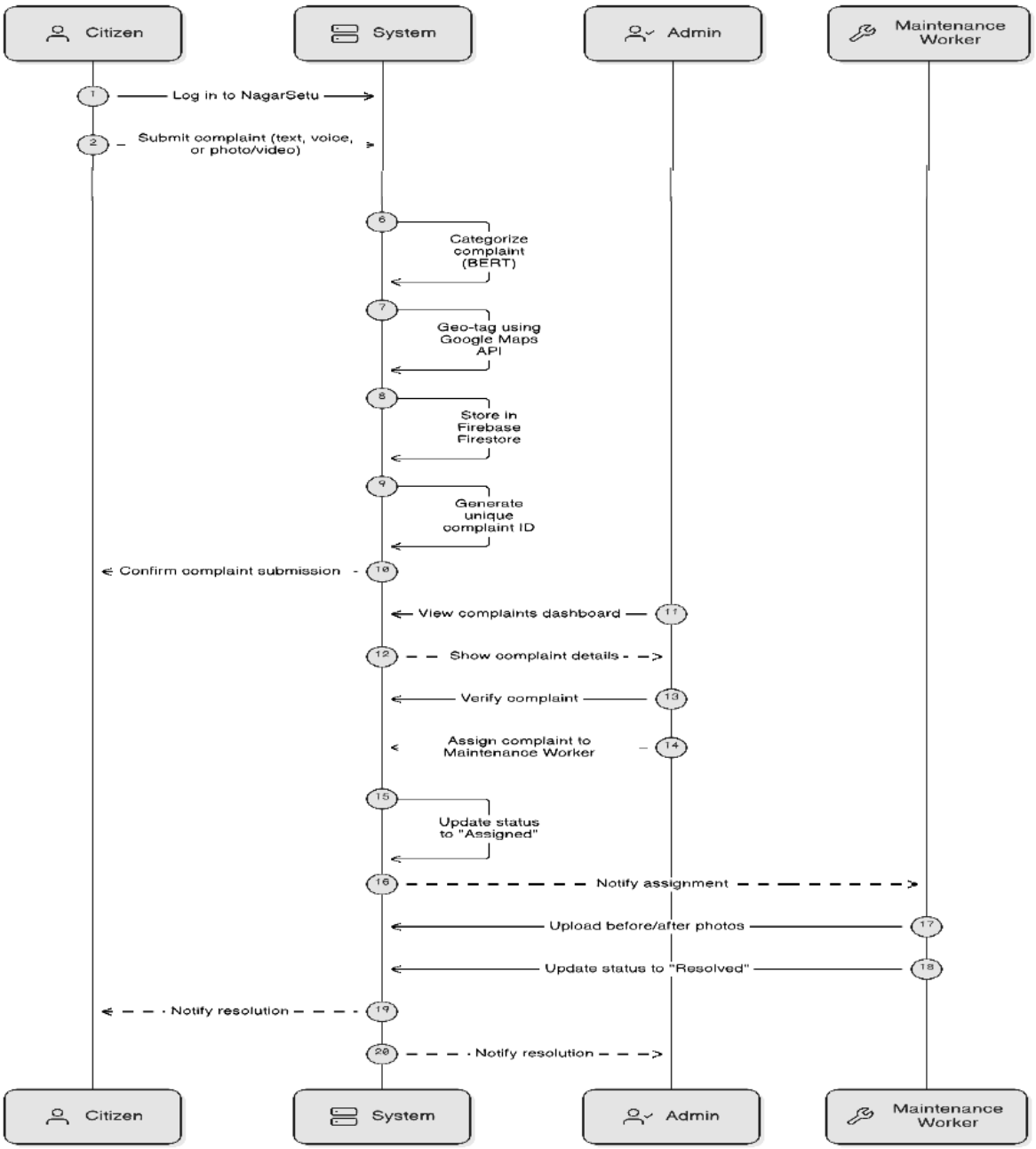


Figure 5.6 Sequence Diagram for NagarSetu

The sequence diagram illustrates the workflow of the MedIntel system, focusing on the interactions between users and the system. It shows the steps involved in user registration, login, document creation and approval, and data retrieval. The diagram highlights the roles of different users (Admin, Super User, Approver, Reviewer, Writer) and their responsibilities in the process. It also shows the flow of data and messages between the users and the system components.

Activity Diagram:

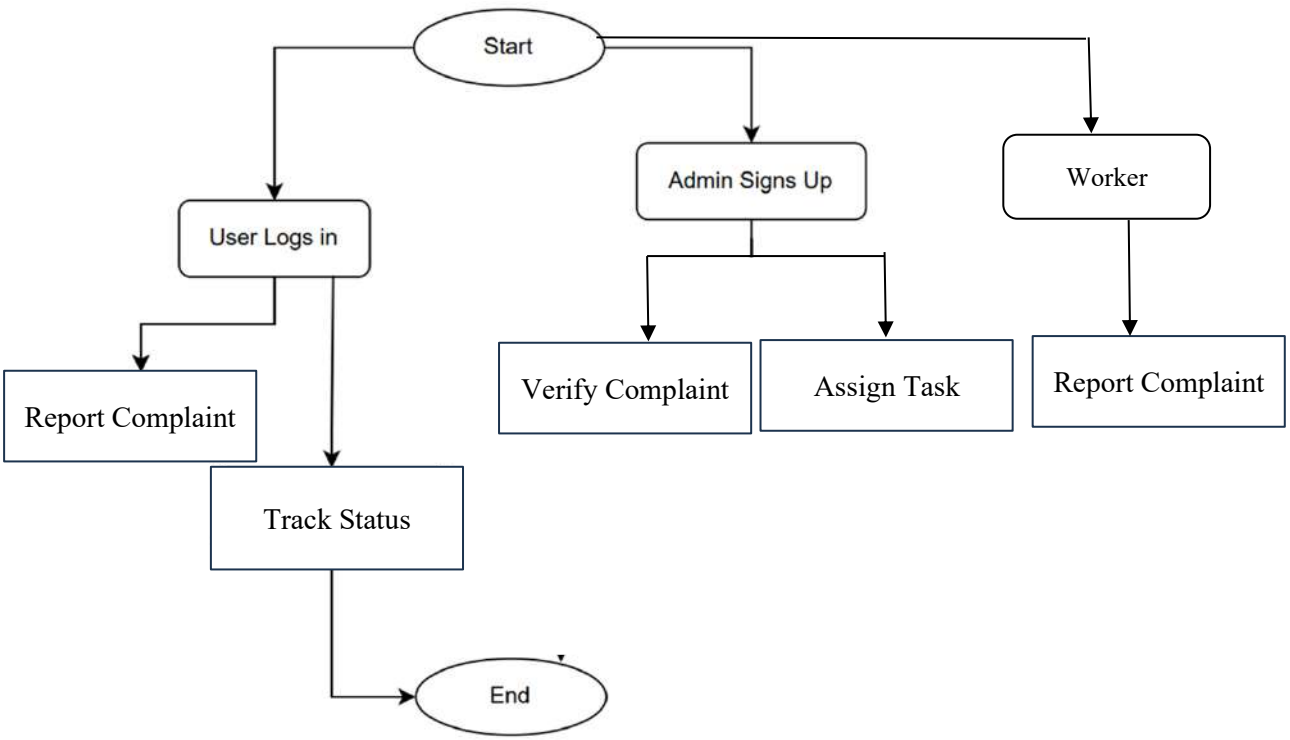


Figure 5.7 Activity Diagram for NagarSetu

The NagarSetu activity workflow begins with a citizen logging into the app or website and choosing to report a complaint .Voice complaints are converted to text using Whisper STT, while images is classified using a CNN model. The system categorizes the complaint using BERT and geo-tags the location with Google Maps API before storing it in Firebase Firestore and generating a unique ID. The admin then logs in, reviews and verifies the complaint, and assigns it to a maintenance worker based on priority and availability. The worker resolves the issue, uploads before-and-after images, and updates the status to “Resolved,”.

Experimental Setup

Software Requirements: -

1.Backend

- Framework: FastAPI – for creating RESTful APIs to handle complaint submission, RTI drafting, and AI/NLP integration.
- Speech-to-Text (STT): OpenAI Whisper (local) or Google Speech-to-Text for transcribing user voice input.
- AI/NLP Modules:
- Complaint Classification: TensorFlow + CNN (e.g., MobileNet) for image-based complaint identification (e.g., pothole, garbage, leak).
- Text Complaint Extraction: BERT (Transformers) model for classifying complaint type from textual or transcribed voice input.
- RTI Drafting: Retrieval-Augmented Generation (RAG) using a fine-tuned LLM to automatically draft RTI applications.
- Firebase Cloud Messaging (FCM) for push notifications.
- Twilio API for SMS notifications on complaint status.
- Geo-tagging: Google Maps API for location-based tagging of complaints and tracking field officer updates.
- Authentication & Authorization: Firebase Authentication integrated with FastAPI middleware.
- Database: Firebase Firestore for storing user complaints, RTI drafts, user details, and officer updates.
- Deployment: Uvicorn server with auto-reload for development and Dockerized FastAPI for production.

2. Frontend

- Framework: React.js with Vite for fast development and bundling.
- Styling: Tailwind CSS for responsive and modern UI.
- REST API from FastAPI backend

3. AI & ML Integration

- Frameworks: TensorFlow, PyTorch (optional), Transformers (Hugging Face).
- Models:
- MobileNet: For image classification of civic issues.
- BERT or DistilBERT: For NLP complaint categorization and RTI drafting.
- Whisper: For voice-to-text complaint input.
- Data Handling: NumPy, Pandas for preprocessing, OpenCV for image transformations.
- Training Environment: Google Colab for dataset training and model evaluation before deployment.

Hardware Requirements: -

1. **Processor:** Minimum of Intel Core i5 or equivalent (Quad-core) for local development and processing.
2. **RAM:** At least 16 GB for smooth running of applications, especially when handling large datasets for report generation.
3. **Storage:** Minimum of 256 GB SSD for fast read/write operations and storage of intermediate data, models, and processed files.

Operating Environment:

4. **Operating System:** Windows 10/11, macOS, or Ubuntu 22.04 (preferred for Python & FastAPI).
5. **Python Version:** 3.9 or higher.
6. **Node.js Version:** 18+ for frontend.
7. **Database Environment:** Firebase Firestore (NoSQL).

Machine Learning & NLP Libraries:

7. **TensorFlow:** For building and running machine learning models.
8. **NLP:** Transformers (BERT), SpaCy, NLTK
9. **Speech:** Whisper, SpeechRecognition
10. **Google Cloud/Colab:** For scaling machine learning experiments and large dataset processing

Chapter 7

Project Planning

Gantt Chart:

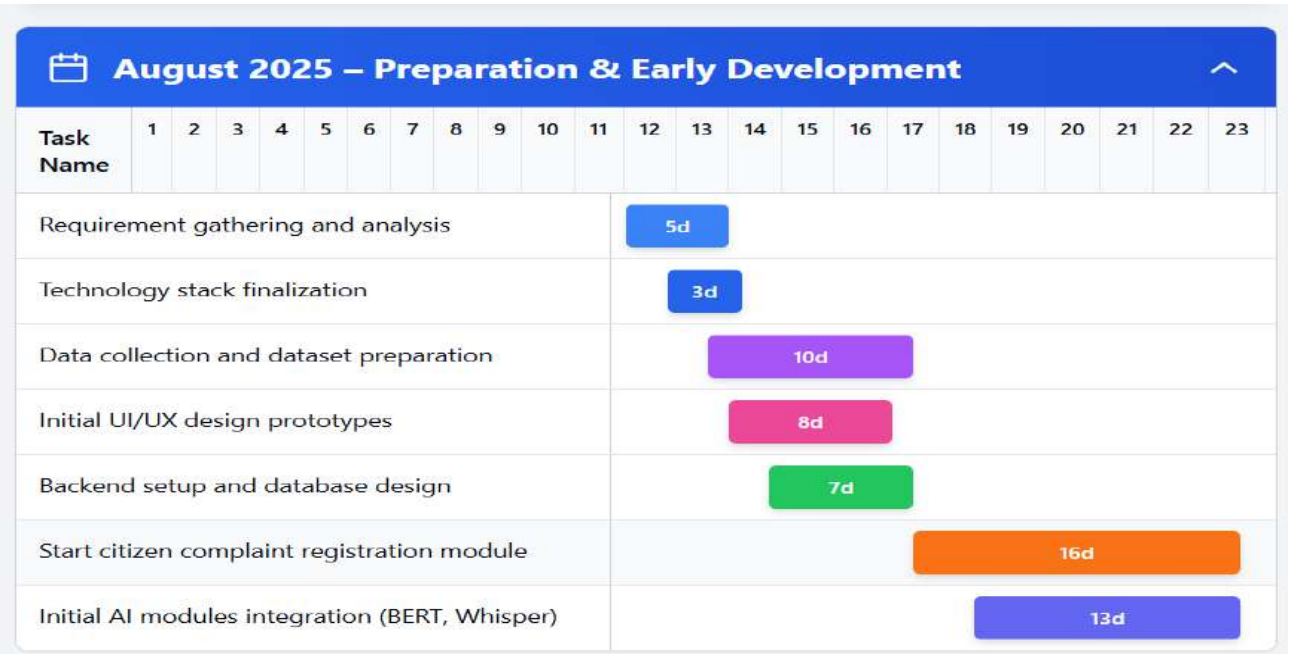


Figure 6.1 August (Gantt Chart)

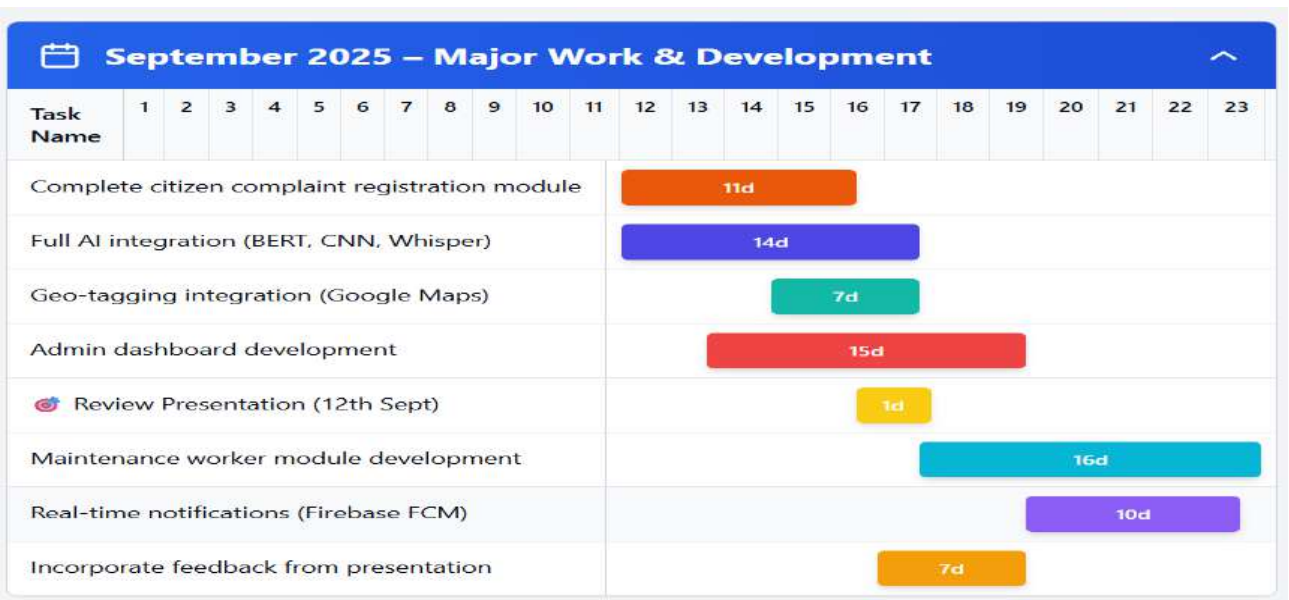


Figure 6.2 September (Gantt Chart)

Figure 6. November (Gantt Chart)

Chapter 8

Expected Outcome

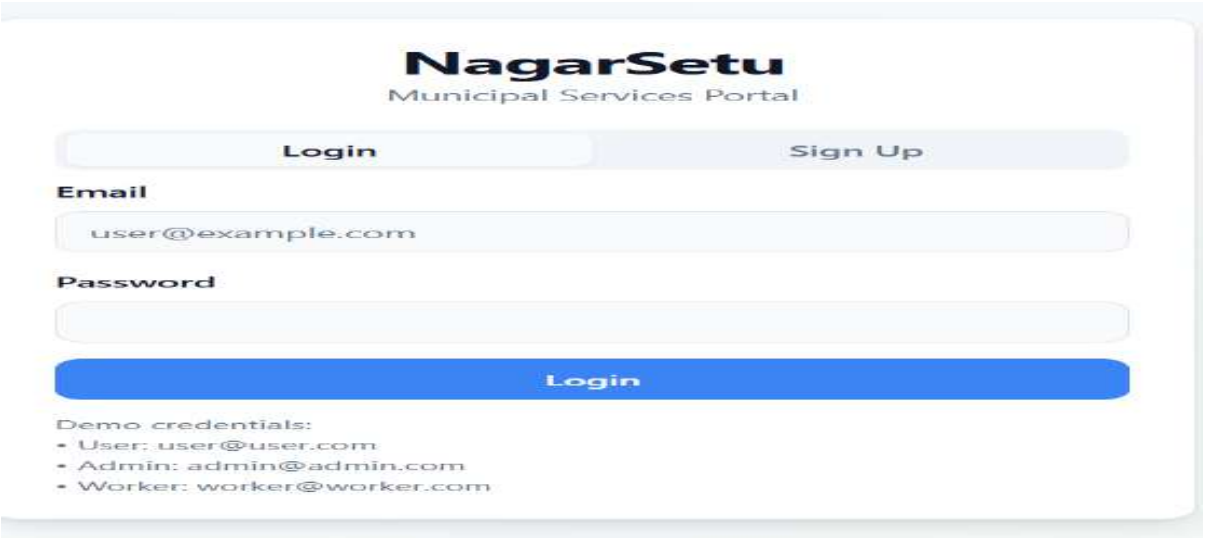


Fig 7.1 Login Page

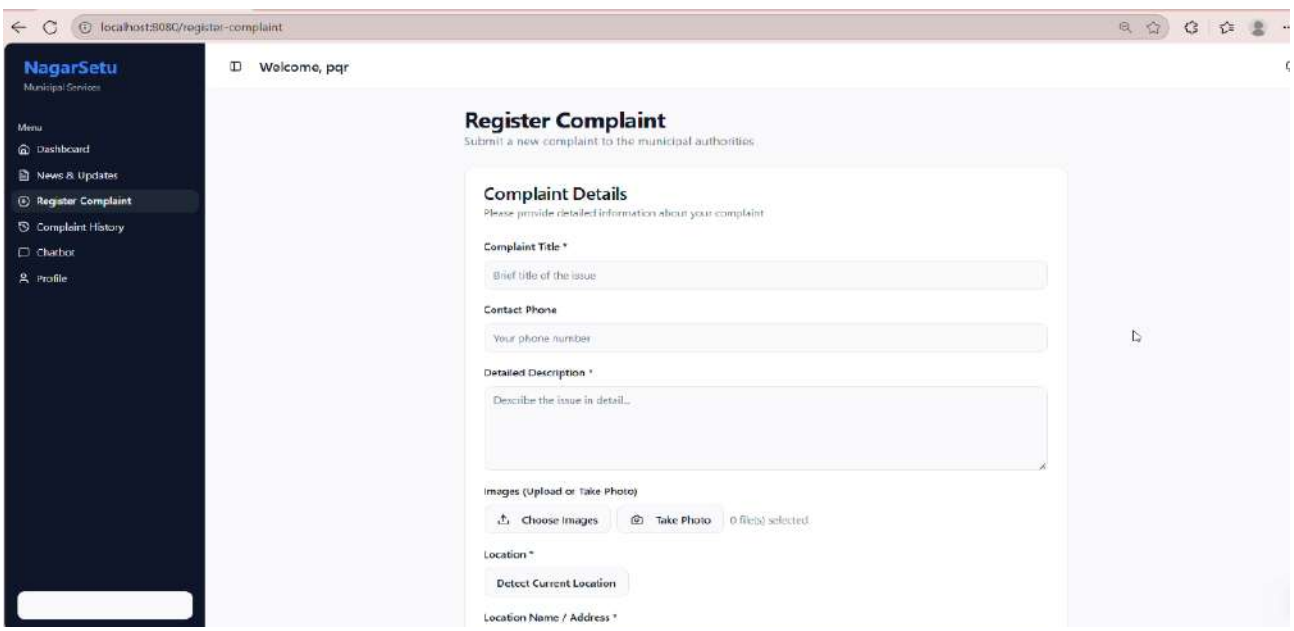
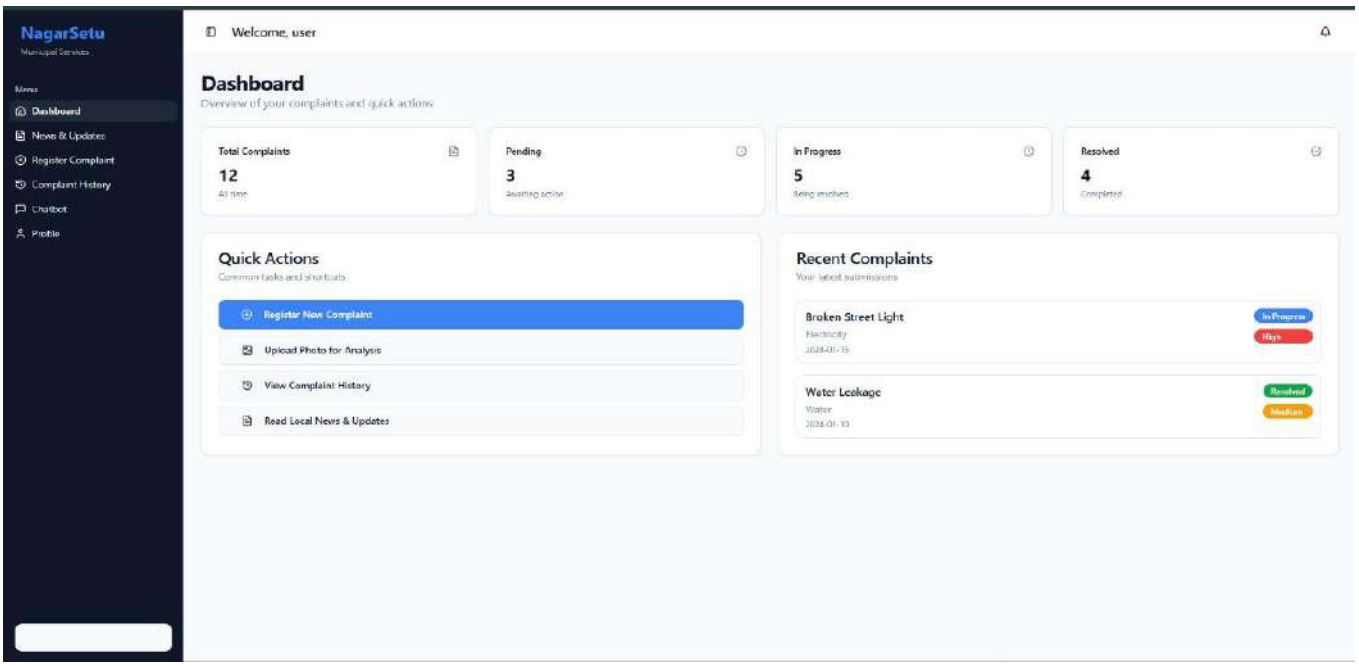


Fig 7. Register Complaint



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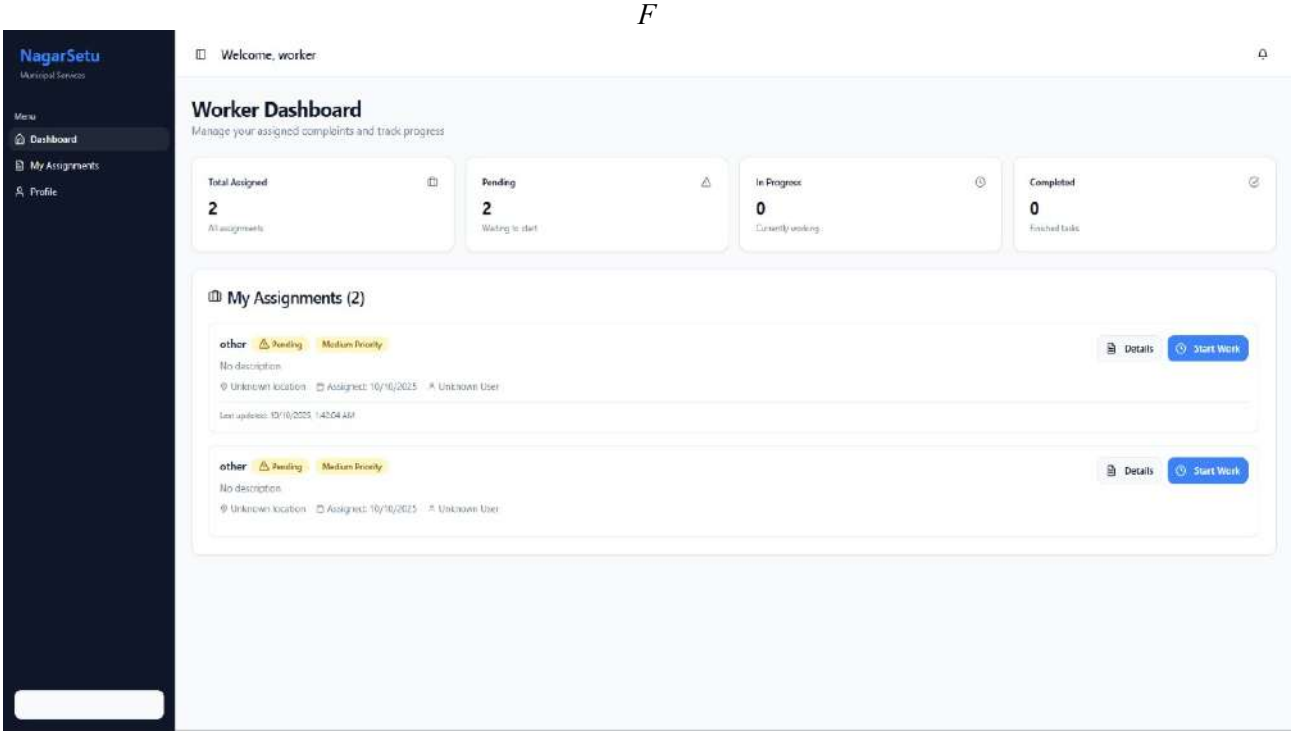

ig 7.1 User Dashbord

Fig 7. Worker Dashboard

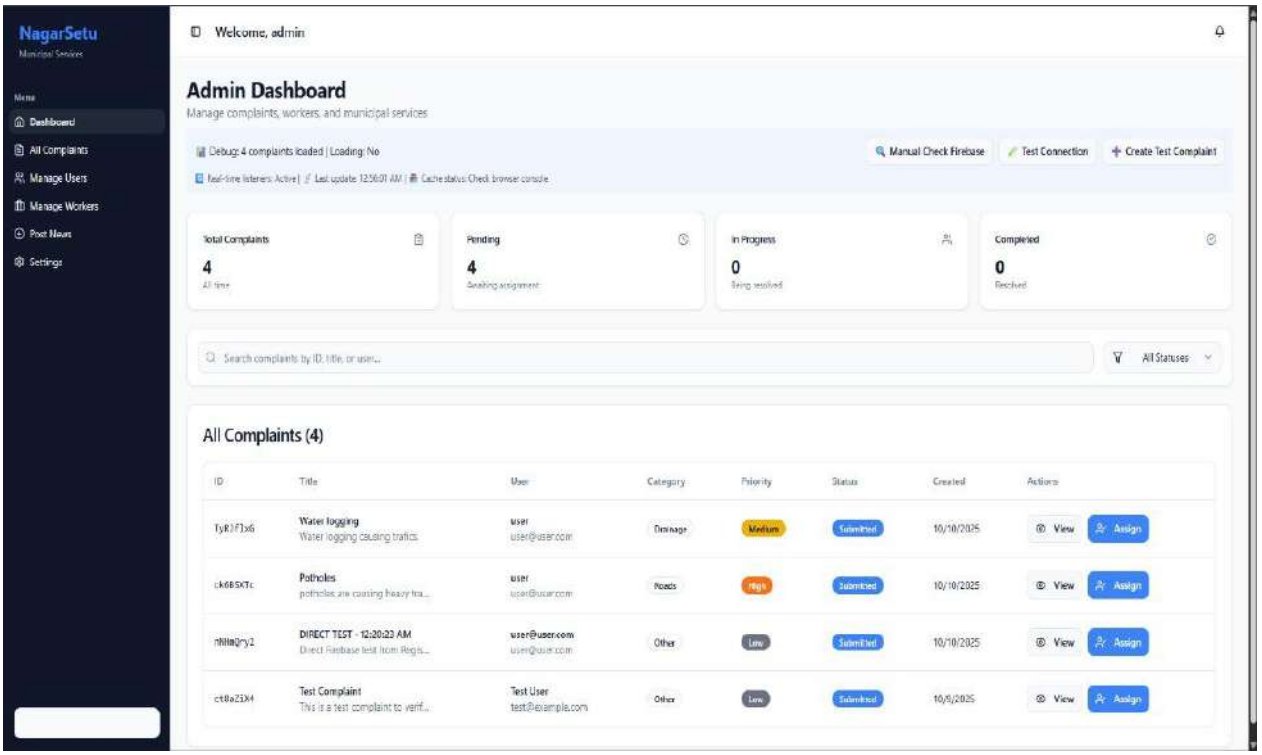

Fig 7. Admin Dahboard

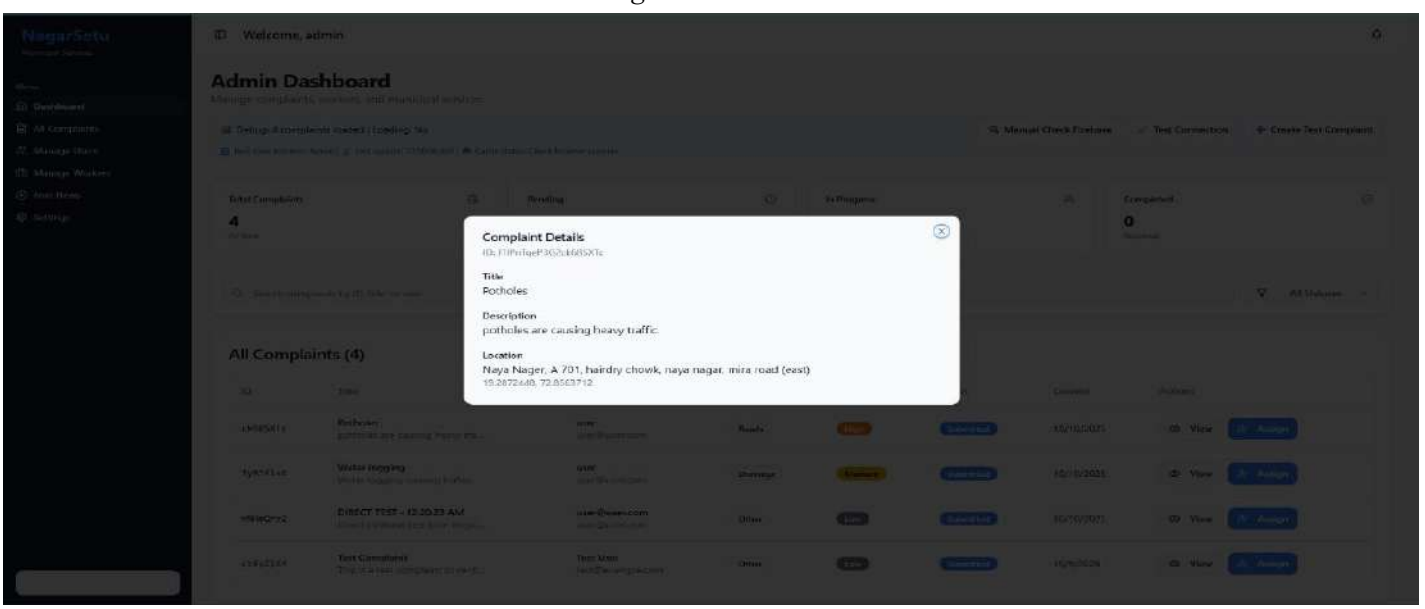

Fig 7. Complaint Details

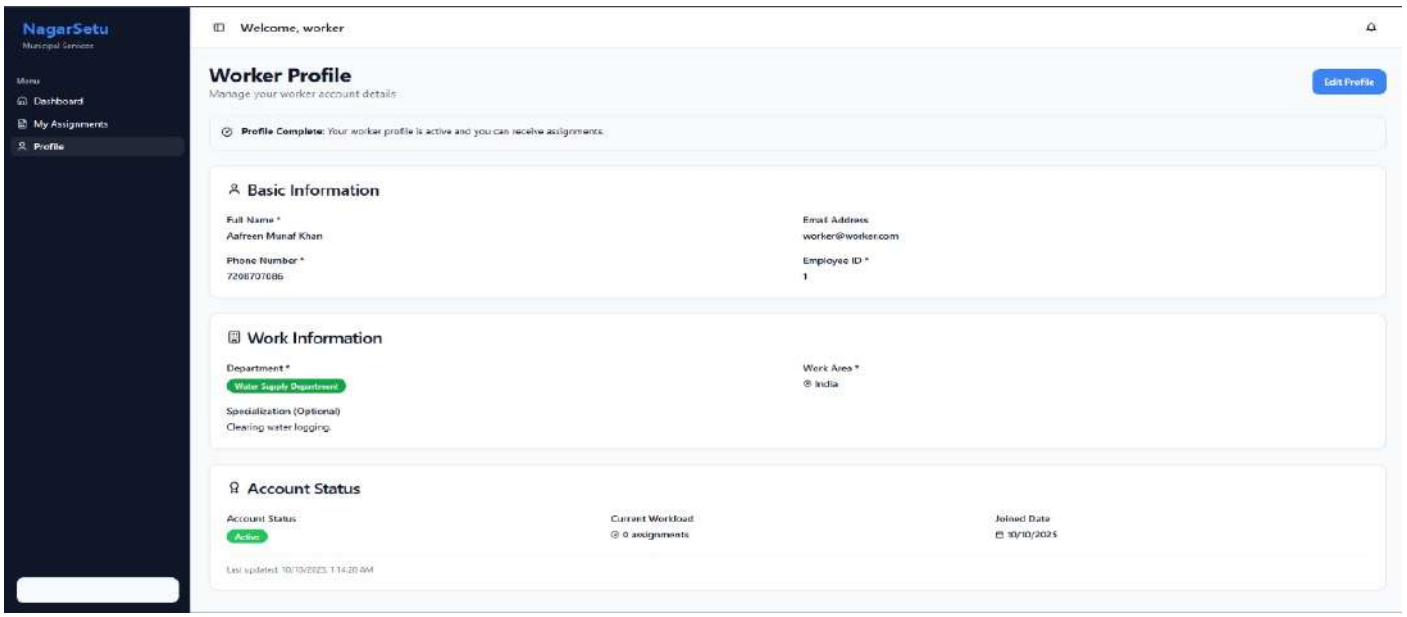

Fig 7. Worker Profile

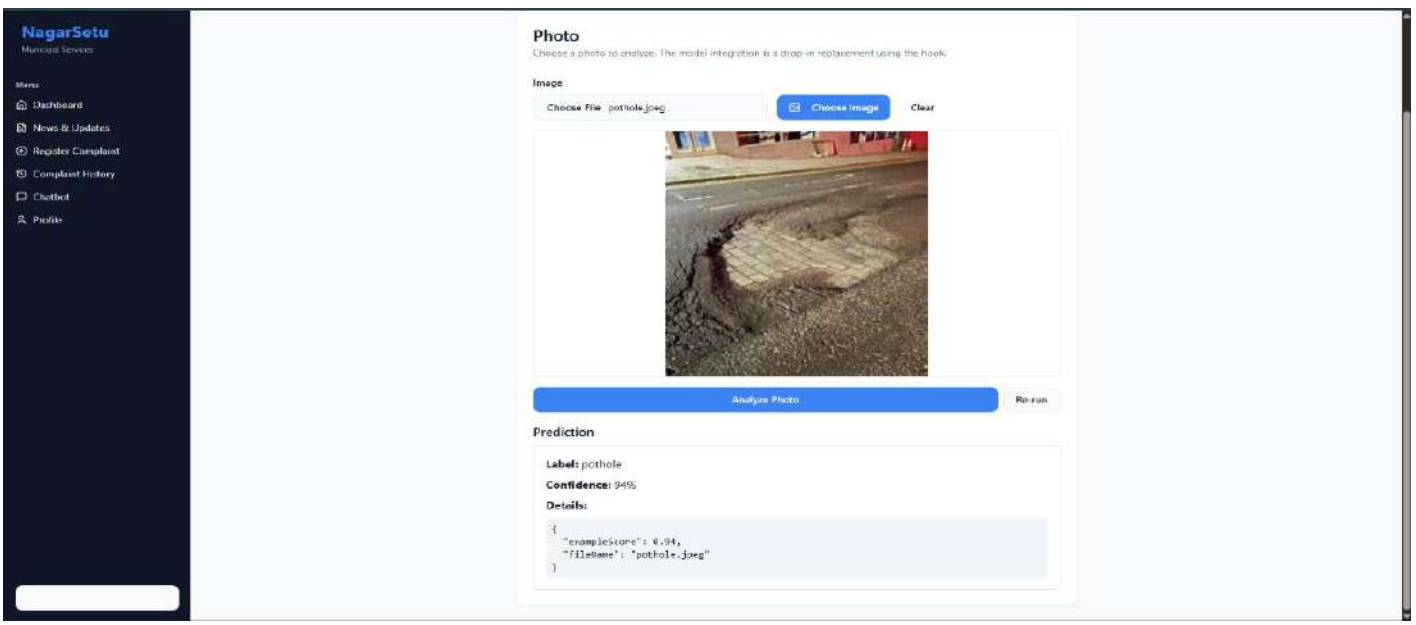

Fig 7. Complaint Location

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